

Training-Free Open-Vocabulary Dense Mapping via Exchanged Semantic–Geometric Verifications

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Abstract—We present a training-free system enabling dense metric 3D semantic mapping over long sequences with open-vocabulary queries. The system follows a decoupled architecture that integrates frozen foundation models across geometric and semantic stages: visual–inertial odometry estimates metric poses, a multi-view depth network recovers dense geometry, and voxel-fused vision–language features encode semantics. Upsampled self-supervised features, residual-free dense CLIP decoding, and template-ensembled queries sharpen the per-frame semantics, with 3D smoothing providing further gains. On TartanAir, vegetation segmentation improves IoU from 58.5 to 65.3 and AUC from 0.752 to 0.786 across eleven trajectories. Without re-tuning, the per-frame components also improve 2D performance, with mIoU increasing from 0.469 to 0.526 on RELLIS-3D and IoU from 0.874 to 0.896 on ORFD. To enable information exchange across the decoupled geometric and semantic stages, we introduce a proposal-verification mechanism. Pose and depth priors propose pixel correspondences that semantic features verify, recovering part of the depth accuracy lost under pose drift. Reused semantic features propose revisits that geometry verifies, increasing valid loop closures and reducing ATE by 15–52% on the five EuRoC-MH sequences. Finally, we adopt a completion-style IoU metric and factor it into labeling and coverage terms, penalizing unreconstructed regions while exposing their respective contributions to 3D semantic map quality.

I. INTRODUCTION

A robot queried for “the gravel path” or “a place to sit” requires a map that is both metric, to support distance-based action, and language-indexed, to support natural-language interaction. Vision–language foundation models provide the latter capability without task-specific training: their features align images with arbitrary text and can, in principle, be attached to any reconstruction [1], [2], [3]. The central systems question is how this semantic representation should interact with established geometric estimation, as this choice determines many properties of the resulting map (Fig. 1).

One approach tightly couples semantics and geometry by incorporating semantic features directly into pose estimation, for example through semantic reprojection residuals [4], feature-metric bundle adjustment [5], or foundation-feature odometry [6]. Although a unified objective is attractive, it couples failure modes: monocular formulations lose metric scale, backend complexity grows with sequence length, and errors in one term propagate to other estimated quantities.

An alternative is modular composition. A classical visual–inertial estimator [7] provides poses with IMU-observable metric scale and bounded local error; a feed-forward multi-view network [8] predicts dense depth corrected by cross-frame pixel correspondences [9]; and frozen self-supervised and vision–language models [10], [11] provide semantic

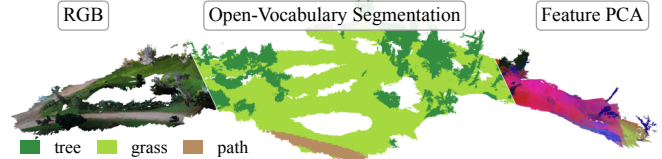


Fig. 1. **System Output.** A fused park reconstruction visualized as RGB, open-vocabulary segmentation for three queries, and principal components of the voxel features.

features. This design isolates failure modes and permits component replacement: stronger frozen models can be introduced without retraining, whereas a trained open-vocabulary segmenter [12], [13] retains the biases of its training distribution at deployment. However, composition limits the information shared across stages. Each stage has cases where its own cues become unreliable: snow and repetitive foliage weaken the detector matches used for depth-scale correction, while appearance-based place recognition cannot reliably distinguish true revisits from perceptual aliasing with a fixed similarity threshold.

We show that modular composition can share information across stages without coupling the estimators. Cross-stage information enters at two interfaces as *proposals verified by another stage*. First, pose and depth priors propose pixel correspondences that frozen features verify, augmenting depth-scale optimization when detector matching fails. Second, reused keyframe features propose revisits that geometry verifies before pose-graph integration. Thus, incorrect proposals are discarded rather than incorporated into the estimate. Across our experiments, verified proposals are effective when introduced into the stage containing the corresponding error.

Evaluation presents a further challenge: online maps lack an aligned reference surface and are often evaluated only on reconstructed points, leaving missing geometry unpenalized. We instead evaluate over the full scene voxels and factor completion IoU into labeling and coverage terms, attributing map quality to semantic labeling and scene coverage.

Contributions. (1) We present a training-free open-vocabulary 3D mapping system integrating frozen foundation models and improve its semantic branch on 2D and 3D benchmarks. (2) We introduce two verified semantic–geometric exchanges and quantify their effects on the corresponding estimation stages. (3) We factor completion-style IoU into labeling and coverage terms, separating semantic quality from incomplete reconstruction.

II. SYSTEM

Figure 2 summarizes the pipeline from sensor input to the queryable map; this section describes each stage in sequence.

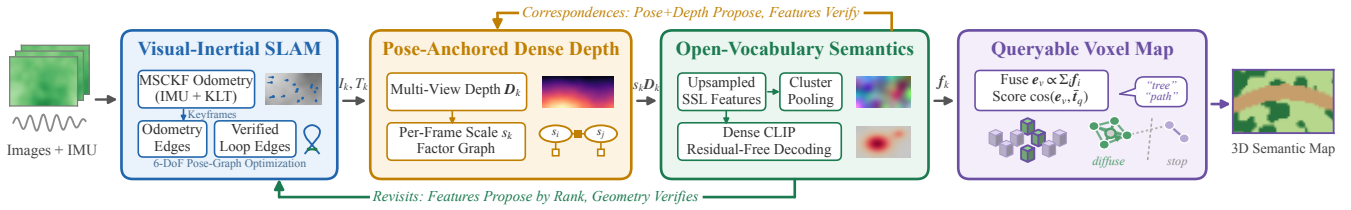


Fig. 2. **System Overview.** A visual-inertial stream passes through SLAM, pose-anchored dense depth estimation, and frozen open-vocabulary semantic encoding to produce a queryable voxel map. Voxel scores are diffused over a feature-gated voxel graph (right), propagating consensus within objects while respecting boundaries. In the depth stage (orange block), per-frame scales are optimized using pixel matches and unit-scale priors. Cross-stage information enters through proposal verification: anchored correspondences (orange arrow) help correct depth scales, while rank-gated revisits (green arrow) update the SLAM pose graph.

A. From Sensor Measurements to Metric Poses

Images and IMU measurements are processed by an MSCKF-based visual-inertial estimator [7]. The estimator tracks sparse corners with KLT, outputs camera poses T_k , and selects keyframes I_k ; verified loop constraints are incorporated into a pose graph over these keyframes (Sec. II-E). Because the IMU observes gravity and scale, the metric scale of each motion increment is continually corrected. Drift accumulates primarily in position and yaw, the unobservable directions, while local scale remains close to the true value.

B. From Poses to Metric Dense Geometry

Keyframes are grouped into submaps and processed by a frozen multi-view depth network [8], which predicts depth maps D_k with accurate structure but uncertain per-frame scale. Following ScaRF-SLAM [9], we recover these scales by optimization on a factor graph [14]. Its variables are the per-frame scalars s_k , and its factors encode pixel correspondences between neighboring frames (Fig. 2). For a correspondence (p, p') between frames i and j , both back-projections must represent the same world point, yielding

$$\mathbf{r}_{ij} = \mathbf{T}_i \pi^{-1}(\mathbf{p}, s_i D_i(\mathbf{p})) - \mathbf{T}_j \pi^{-1}(\mathbf{p}', s_j D_j(\mathbf{p}')), \quad (1)$$

which is linear in the two scale variables because the poses remain fixed. A unit-scale prior on each variable and a tight prior on one anchor frame complete the graph. The resulting objective minimizes $\sum \rho(\|\mathbf{r}_{ij}\|) + \lambda \sum_k (s_k - 1)^2$ under a Huber kernel ρ [15]. Anchoring depth to the poses transfers the IMU-derived metric scale to the reconstruction.

Correspondences are obtained from two sources. A detector-based matcher [16], [17] provides correspondences where sufficient texture is available. When detector matches are sparse, the first geometry-semantic exchange provides additional candidates. A uniform pixel grid is reprojected using the pose and depth prior:

$$\hat{\mathbf{p}} = \pi(\mathbf{T}_{ji} \pi^{-1}(\mathbf{p}, D_i(\mathbf{p}))), \quad (2)$$

thereby proposing detector-free matches that frozen features only need to accept or reject. Each proposal passes three gates: occlusion, $|D_j(\hat{\mathbf{p}}) - \hat{z}_j|/D_j(\hat{\mathbf{p}}) < \tau_o$, where $\hat{z}_j$ is the propagated depth; semantic agreement, $\langle \mathbf{f}_i(\mathbf{p}), \mathbf{f}_j(\hat{\mathbf{p}}) \rangle \geq \tau_c$, for frozen features $\mathbf{f}$; and distinctiveness. The final gate rejects a proposal if any location in a small window around $\hat{\mathbf{p}}$ provides a clearly stronger match, rather than relocating the correspondence. Accepted proposals enter (1) only for

frame pairs with fewer than a minimum number of detector matches, and their number is capped per pair.

C. From Keyframes to Pixel-Level Semantics

Each keyframe is embedded by two complementary branches. A self-supervised backbone [10] with an implicit upsampler [18] generates a dense feature map whose higher resolution better preserves object boundaries blurred by patch-level features. Clustering these features partitions the image into boundary-aligned segments. In parallel, dense CLIP decoding produces language-aligned features. Following ClearCLIP [19], we evaluate the final Transformer block using attention alone, discarding its residual and feed-forward paths. Since CLIP’s objective aligns only the pooled token with text, this modification improves language alignment at the patch level. Pooling language features within each segment yields per-pixel embeddings $\mathbf{f}_k$ that combine segment-level coherence with open-vocabulary alignment.

For a text query q , we average its unit-normalized embeddings across the standard 80 prompt templates [11]:

$$\bar{\mathbf{t}}_q = \frac{1}{80} \sum_m E_{\text{text}}(\text{template}_m(q)). \quad (3)$$

This design better matches the text encoder’s training distribution, where textual inputs are typically richer than isolated words, while reducing sensitivity to template-specific embedding components. Template ensembling also reduces query-dependent similarity offsets, enabling a common decision threshold across queries.

D. Fusion into a Queryable Voxel Map

Pixels on the shared feature grid are back-projected using $s_k D_k$ and T_k and accumulated into voxels. Each voxel fuses its observations by averaging their embeddings and normalizing the result, $\mathbf{e}_v = \sum_{i \in v} \mathbf{f}_i / \|\sum_{i \in v} \mathbf{f}_i\|$. The resulting direction represents the observation consensus; optional view-quality weighting can also be applied. Each query scores all voxels by cosine similarity to $\bar{\mathbf{t}}_q$, contrasted against negative prompts.

Voxels observed from few or misregistered views may receive outlier scores. Before thresholding, we therefore diffuse the score field $\mathbf{s}$ over a feature-gated voxel graph. Voxels within radius r are connected with weight $w_{ij} = \exp(-(1 - \cos(\mathbf{e}_i, \mathbf{e}_j))/\tau)$, and two Jacobi iterations,

$$\mathbf{s} \leftarrow \mathbf{D}^{-1}(\mathbf{s} + \mathbf{W}\mathbf{s}), \quad \mathbf{D} = \mathbf{I} + \text{diag}(\mathbf{W}\mathbf{1}), \quad (4)$$



Fig. 3. **Query-Time Smoothing.** Errors before smoothing on TartanAir P002 (left) and voxels whose predictions change after smoothing (right).

replace each score with a weighted neighborhood average. Because fused embeddings differ across object boundaries, graph edges weaken at these boundaries; consensus therefore propagates within objects while remaining localized (Fig. 3). The graph depends only on the map and is built once during voxelization, while diffusion acts on the scalar score field, leaving the stored embeddings unchanged and allowing each query to operate on the same map.

E. Loop Closure Using the Map’s Features

The second exchange feeds information from the map back to pose estimation. For each keyframe, the mean of its upsampled feature grid is reused as a retrieval descriptor $\mathbf{d}_i$, requiring neither an additional model nor another forward pass. In self-similar environments, absolute similarity cannot reliably distinguish revisits from perceptual aliasing, precluding a fixed threshold; rank, however, remains discriminative. A pair (i, j) is proposed only if it satisfies mutual top-1 ranking and temporal separation:

$$j = \arg \max_{k: |t_i - t_k| > \tau_t} \mathbf{d}_i^\top \mathbf{d}_k, \quad i = \arg \max_{k: |t_j - t_k| > \tau_t} \mathbf{d}_j^\top \mathbf{d}_k, \quad (5)$$

a criterion that does not require metric poses, which may itself be corrupted by drift. Each candidate is then verified geometrically. Matches from frame i and its stereo pair are triangulated to obtain metric structure, and PnP-RANSAC [20], [21] against frame j recovers a relative pose with sufficient inliers. Verified constraints are incorporated into 6-DoF pose-graph optimization (PGO) with VIO odometry edges and Huber-robust [15] loop edges. The two exchanges are symmetric: geometry proposes correspondences that features verify, whereas features propose revisits that geometry verifies. In both cases, verified proposals enter the stage containing the error: scale optimization for depth error and PGO for drift.

F. Map Evaluation

Let S^* denote the scene support, defined as the voxels occupied by ground-truth depth accumulated at the mapping stride. A voxel in S^* is covered if the map reconstructs geometry within one voxel size. Uncovered positive voxels count as false negatives, and completion IoU is evaluated over S^* . The metric factors as

$$\text{IoU}_c = \underbrace{\frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}_{\text{cov}}}}_{L \text{ (labeling)}} \times \underbrace{\frac{\text{TP} + \text{FP} + \text{FN}_{\text{cov}}}{\text{TP} + \text{FP} + \text{FN}_{\text{cov}} + \text{FN}_{\text{unc}}}}_{\tilde{C} \text{ (coverage)}}, \quad (6)$$

where FN_{cov} denotes covered positive voxels assigned an incorrect label, and FN_{unc} denotes positive voxels that were not reconstructed. The factor L measures labeling quality on covered voxels, whereas $\tilde{C}$ measures coverage; their product preserves the penalty for missing geometry.

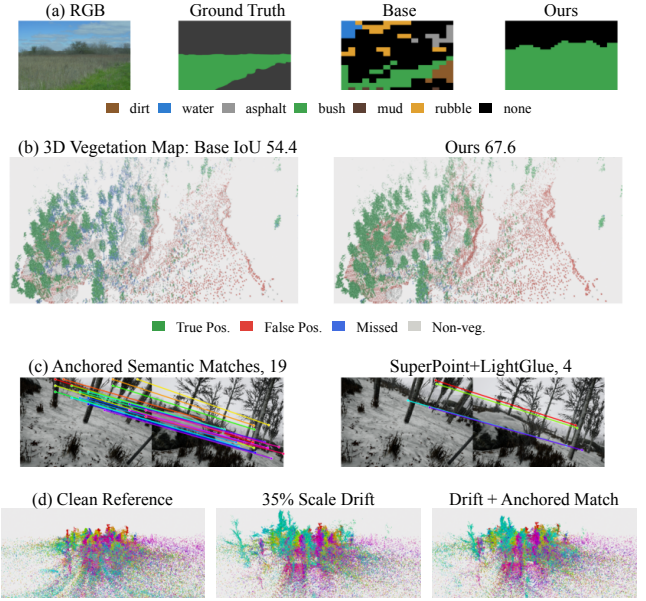


Fig. 4. **Qualitative Results.** (a) 2D segmentation on RELLIS-3D, obtained by an argmax over the six queried classes. (b) 3D vegetation maps on TartanAir P002. (c) Anchored matching compared with detect-and-describe matching on winter frames. (d) Keyframes 62–82 fused with distinct per-keyframe hues: introduced drift misaligns trees, whereas anchored correspondences reduce the misalignment.

III. EXPERIMENTS

We investigate three questions: *Q1: How does the training-free semantic segmentation perform?* *Q2: What does improved geometry contribute to the semantic map?* *Q3: Do the geometry-semantic exchanges improve their target stages?*

Setup. We evaluate semantic mapping on TartanAir vegetation (4 environments, 11 trajectories) and the complete RELLIS-3D [22] and ORFD [23] test sets; geometry sources on a TartanAir V2 trajectory with IMU data; depth-scale correction on TartanAir [24] winter forest sequences (AbsRel below 80 m); and trajectories on the five EuRoC machine-hall sequences [25] using a stereo-inertial front end (SE(3)-aligned ATE and RPE at 1 s). A single configuration is used across benchmarks: smoothing uses $\tau=0.1$, a radius of $2.5 \times$ the voxel size, and two iterations; matching gates require 10% depth agreement and 0.65 feature cosine similarity; loop proposals use $\tau_t=30$ s and require 20 PnP inliers. On one NVIDIA RTX 5090 GPU, semantic requires 22 ms per frame, and geometry requires 0.5–2.1 s per submap chunk.

A. Q1: Quality of Training-Free Semantics

Table I ablates the 3D pipeline across eleven trajectories. We report AUC alongside IoU because fixed thresholds do not transfer reliably across domains. The feature and decoding choices in Sec. II-C account for most of the improvement. Query-time smoothing is compatible with all tested configurations and, when combined with octree pooling, improves AUC on all eleven trajectories; additional feature-space mechanisms overlap with the signal provided by the decoder. For 2D evaluation, the same configuration transfers without re-tuning and improves the base system [26] from

TABLE I

3D SEMANTIC SEGMENTATION ABLATION ON TARTANAIR VEGETATION

3D Configuration	IoU@0.5 $\uparrow$	AUC $\uparrow$
DINOv2 + MaskCLIP (Base) [26]	58.5 $\pm$ 13.7	0.752 $\pm$ 0.071
DINOv3 Vit-Up + ClearCLIP + Templates	64.1 $\pm$ 10.8	0.760 $\pm$ 0.060
+ Query Smoothing (Ours)	65.3$\pm$10.9	0.786 $\pm$ 0.058
Octree Pooling (Homogeneity, on Vit-Up)	60.4 $\pm$ 15.0	0.791 $\pm$ 0.068
+ Query Smoothing (Ours)	61.1 $\pm$ 15.1	0.799$\pm$0.069
Full Stack + Octree	64.9 $\pm$ 10.7	0.786 $\pm$ 0.062

Mean $\pm$ std over 11 trajectories; AUC: threshold-free ranking. TA: TartanAir veg.

TABLE II

GEOMETRY SOURCES UNDER DIFFERENT EVALUATION PROTOCOLS

Geometry	Self IoU (AUC)	Common IoU (AUC)	L	$\tilde{C}$	Compl.
Ground Truth	74.3 (0.78)	48.4 (0.66)	76.5	60.7	46.4
VGGT [27]	56.2 (0.72)	50.1 (0.64)	69.9	12.1	8.5
lingbot [28]	64.4 (0.74)	48.9 (0.66)	68.5	22.4	15.3
ScaRF [9]	71.0 (0.80)	51.9 (0.71)	72.4	37.4	27.1

TartanAir V2 Gascola, same semantics/frames; ScaRF on mono-VIO poses.

Self: own points; Common: co-covered regions; L , $\tilde{C}$, Compl. = $L\tilde{C}$: Eq. (6) over S^* .

0.469 to 0.526 mIoU on RELIS-3D and from 0.874 to 0.896 IoU on ORFD.

B. Q2: Coverage Dominates Geometry Comparisons

Table II evaluates one semantic pipeline with four geometry sources under three protocols. Self-evaluation favors sparse maps by scoring only reconstructed regions, where misplaced voxels inherit labels from the nearest annotated point. Restricting evaluation to regions covered by all maps similarly excludes difficult scene regions. Completion IoU instead penalizes each map for unreconstructed voxels, and its factorization identifies the source of the difference. The labeling factors L are close, as maps with lower coverage are evaluated on easier subsets; nevertheless, among the estimated sources, our decoupled ScaRF geometry [9] achieves the highest L , indicating better 3D semantic map quality. The coverage factors $\tilde{C}$ vary substantially, with ScaRF again highest among them, indicating more complete reconstruction. Together, these gains yield the highest completion-style IoU of the estimated sources.

C. Q3: Exchanges Improve Their Target Stages

Depth Scale. On a winter forest sequence, 78% of neighboring keyframe pairs fall below the 60-match threshold that activates the exchange. The anchored proposals improve upon SuperPoint+LightGlue [16], [17] in pixel-match error (median 1.1 vs. 1.4%; p95 79 vs. 99%). Under 35% synthetic scale drift, the exchange recovers approximately one-fifth of the lost depth accuracy (AbsRel 0.429 $\rightarrow$ 0.393, relative to a clean reference of 0.244); on clean sequences, its performance remains within 0.003 of the baseline. Three controls isolate the mechanism. First, the pose anchor is most influential: removing it biases the recovered scales by -20.9% . Second, the matcher is not the limiting component: twelve alternatives spanning sparse, dense, and recent learned designs remain within one percent of the default matcher on clean sequences, and the strongest recovers less than half of the exchange’s improvement under drift while

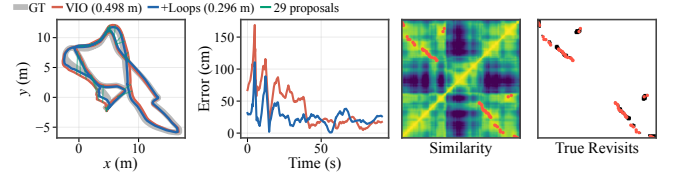


Fig. 5. **Semantic loop closure on MH.05**, for which DBoW detections degrade the trajectory. From left to right: 29 rank-gated proposals (green chords) and the corrected trajectory (blue) following ground truth; position error over time; keyframe-descriptor similarity; and ground-truth revisit structure (within 2 m and 45° , with more than 30 s of separation). Proposals (red) are concentrated in the true-revisit blocks.

TABLE III

LOOP CLOSURE ON EUROC (ATE; RPE@1s TRANS. / ROT.)

Seq.	Run	Loops	ATE (m) $\downarrow$	RPE@1s (cm / $^\circ$) $\downarrow$
MH.01	OpenVINS	—	0.112	4.1 / 0.24
	DBoW	0	0.112	4.1 / 0.24
	+Semantic	30/30	0.095 (−15%)	4.0 / 0.23
MH.02	OpenVINS	—	0.310	9.1 / 0.44
	DBoW	0	0.310	9.1 / 0.44
	+Semantic	15/17	0.255 (−18%)	8.7 / 0.40
MH.03	OpenVINS	—	0.139	6.4 / 0.23
	DBoW	4	0.128 (−8%)	6.7 / 0.20
	+Semantic	52/56	0.107 (−23%)	5.9 / 0.23
MH.04	OpenVINS	—	0.194	6.2 / 0.32
	DBoW	21	0.109	6.3 / 0.24
	+Semantic	21/21	0.093 (−52%)	6.0 / 0.32
MH.05	OpenVINS	—	0.498	18.9 / 0.98
	DBoW	14	0.526 (+6%)	19.1 / 0.93
	+Semantic	27/29	0.296 (−41%)	17.5 / 1.08

+Semantic: verified rank-gated proposals incorporated into PGO.

requiring 6–22 $\times$ more matching time. Third, rejection is preferable to relocation: assigning verified proposals to the strongest nearby feature becomes beneficial only between 10% and 35% synthetic drift, above the drift produced by our visual-inertial front end, and degrades clean-sequence performance. The drift structure, rather than its magnitude, appears to determine this regime. Drift with bounded local scale leaves depth accuracy unchanged at both tested magnitudes, whereas cumulative monocular-style scale error reproduces both the deficit and its correction (Fig. 4c,d).

Pose Graph. Table III and Fig. 5 shows the loop-closure performance. On MH.01/02, bag-of-words retrieval [29] detects no loops and therefore reproduces the odometry trajectory. In contrast, the rank gate proposes 30/17 pairs, of which 30/15 pass verification (ATE -15% / -18% ; RPE marginally better than VIO). On MH.05, where DBoW detections degrade ATE by $+6\%$, our method reduces it by 41%. DBoW is precise but identifies few true revisits, whereas the rank gate covers 1.6–2.5 $\times$ more of the true revisits at 77–87% raw proposal precision; geometric verification rejects the remaining false proposals.

IV. CONCLUSION

Composing frozen models isolates failure modes, while verified cross-stage proposals restore useful information exchange where individual stages are weak. Factoring completion-style IoU into labeling and coverage clarifies their contributions to 3D map quality. A parkland dataset and robot deployment are under development.

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